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The system of equations,

$$\begin{aligned} -3x - 2y &= 3 \\ x - y &= -6 \\ -2x - 2y &= -5 \end{aligned}$$

where $x, y \in \mathbb{R}$, has no solution.

Find the vector $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} -3 \\ 4.111 \end{pmatrix} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded **1** mark. 

You scored **1** mark for this part.

Score: 1/1 

Answered

The system of equations,

$$-6x + 4y = 2$$

$$5x - y = 4$$

$$4x - y = 4$$

where $x, y \in \mathbb{R}$, has no solution.

Find the vector $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \boxed{1.455} \\ \boxed{2.667} \end{pmatrix} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded 1 mark. 

You scored 1 mark for this part.

Score: 1/1 

Answered

The system of equations,

$$\begin{aligned}x - 4y &= -3 \\ 6x - 5y &= -6 \\ x + 2y &= 5\end{aligned}$$

where $x, y \in \mathbb{R}$, has no solution.

Find the vector $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \boxed{0.195} \\ \boxed{1.294} \end{pmatrix} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded 1 mark. 

You scored 1 mark for this part.

Score: 1/1 

Answered

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Find the least squares line of best fit $y = a + bx$ for the data points:

$$\{(-4, -6), (-6, 0), (4, 4), (-5, -1)\}$$

$a =$ 

$b =$ 

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Gap 0

Your answer is correct. You were awarded **1** mark. 

Gap 1

Your answer is correct. You were awarded **1** mark. 

You scored **2** marks for this part.

Score: 2/2 

Answered

Find the least squares line of best fit $y = a + bx$ for the data points:

$$\{(5, 2), (-5, -2), (-4, 3), (-2, 3)\}$$

$$a = \boxed{1.77} \quad \checkmark$$

$$b = \boxed{0.18} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Gap 0

Your answer is correct. You were awarded **1** mark. 

Gap 1

Your answer is correct. You were awarded **1** mark. 

You scored **2** marks for this part.

Score: 2/2 

Answered